

Secure Exit

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ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO5-AT3-Level Exam3

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📖 Level Exam

</> Java

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QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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Level 3 — Hard (H) Permutations — Numbers

✓ Submitted

Write a program to print unique permutations of a given number

Sample Input:

Given Number: 143

Sample Output:

Permutations are:

134

143

314

341

413

431

Testcase: 1. 0

2. 111

3. 505

4. -143

5. -598

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] a:
4         Scanner sc=new Scanner(System.:
5         String num = sc.nextLine();
6         permute(num, "");
7     }
8     static void permute(String s, Stri
9         if (s.length()==0) {
```

```
10         System.out.println(ans);
11         return;
12     }
13     for (int i=0; i<s.length(); i++)
14         char ch = s.charAt(i);
15         if (s.indexOf(ch) != i) {
16             continue;
17         }
18         String remaining = s.substring(i+1, s.length());
19         permute(remaining, ans+ch);
20     }
21 }
22 }
23
```



Output

```
143
134
413
431
314
```

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Level Exam

Java

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QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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Level 3 — Hard (H) Roman — Control

Submitted

Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.

Symbol Value

I 1

V 5

X 10

L 50

C 100

D 500

M 1000

For example, 2 is written as II in Roman numeral, just two ones added together. 12 is written as XII, which is simply X + II. The number 27 is written as XXVII, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:

- I can be placed before V (5) and X (10) to make 4 and 9.
- X can be placed before L (50) and C (100) to make 40 and 90.
- C can be placed before D (500) and M (1000) to make 400 and 900.

Given a roman numeral, convert it to an integer.

Example:

Input: s = "III"

Output: 3

Testcase:1. LVIII

2. MCMXCI
3. V
4. LZAll
5. MCCDTIV

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] a:
4         Scanner sc= new Scanner(System
5         String s= sc.next();
6         int total=0;
7         for (int i=0; i<s.length();i++)
8             int value=0;
9             switch(s.charAt(i)) {
10                 case 'I':value=1;break;
11                 case 'V':value=5;break;
12                 case 'X':value=10;break;
13                 case 'L':value=50;break;
14                 case 'C':value=100;brea
15                 case 'D':value=500;brea
16                 case 'M':value=1000;brea
17             }
18             if(i+1 < s.length()) {
19                 int next=0;
20                 switch(s.charAt(i+1)) {
21                     case 'I':value=1;b
22                     case 'V':value=5;b
23                     case 'X':value=10;b
24                     case 'L':value=50;b
25                     case 'C':value=100,
26                     case 'D':value=500,
27                     case 'M':value=1000
28                 }
29                 if (value<next) {
30                     total-=value;
31                 }
32                 else{
33                     total+=value;
34                 }
35             }
36             else{
37                 total+=value;
```

```
38         }  
39     }  
40     System.out.println(total);  
41 }  
42 }  
43
```

Output

3



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